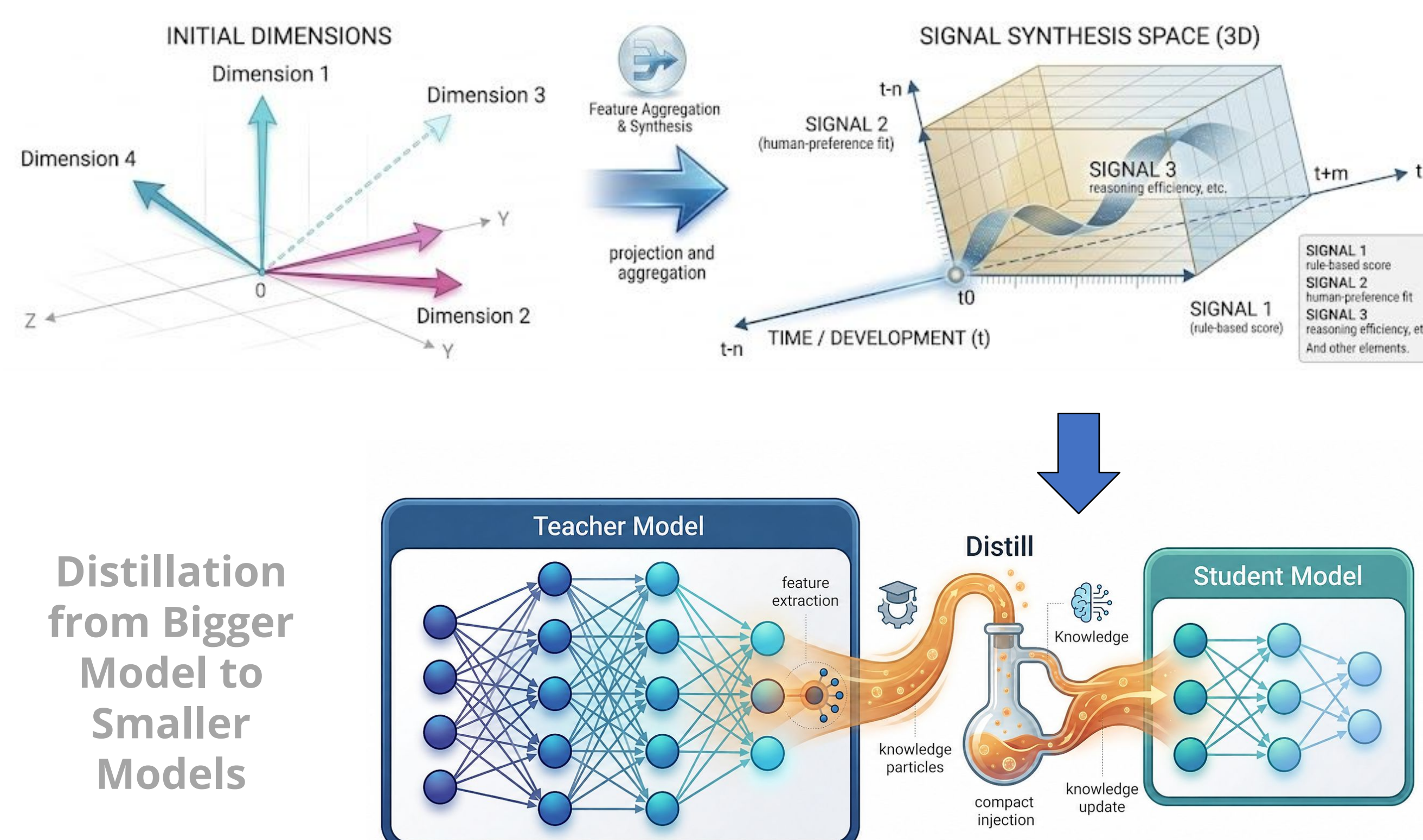
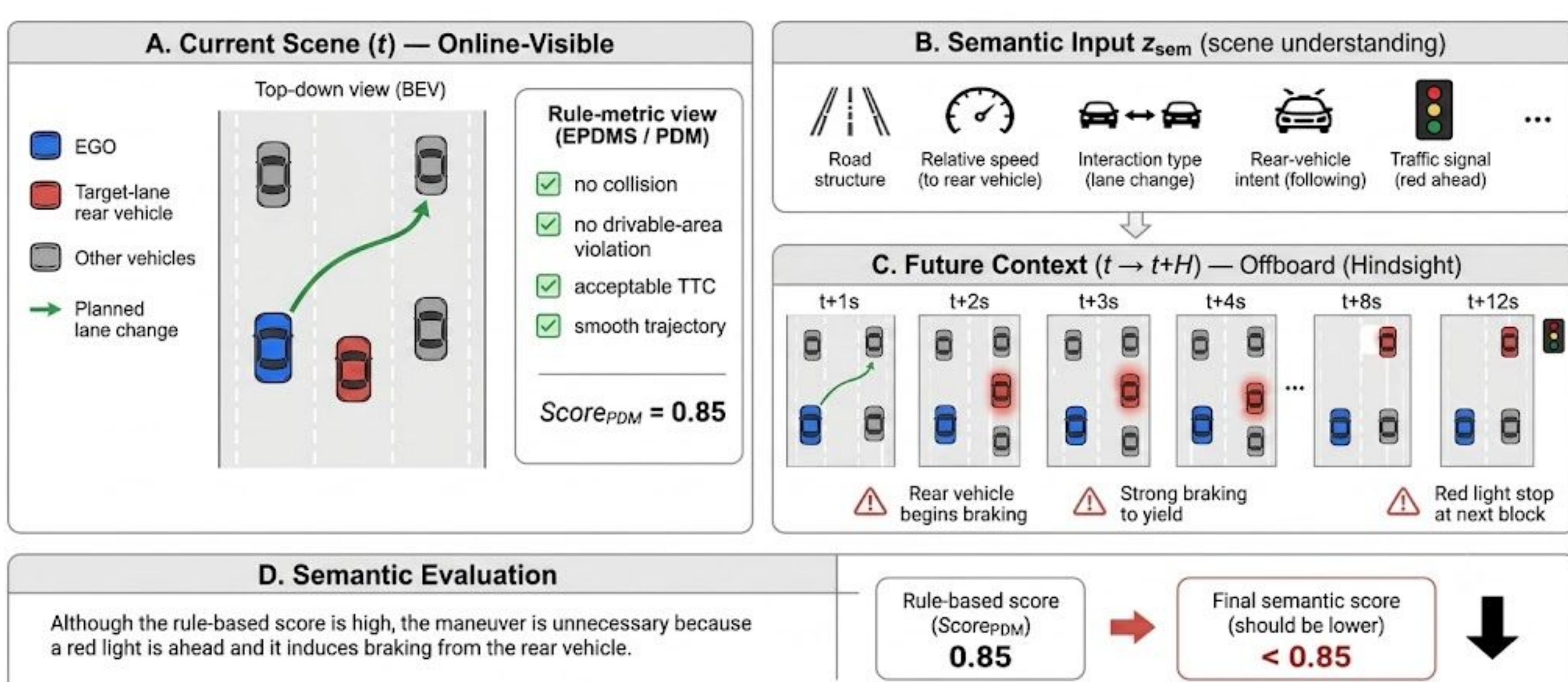
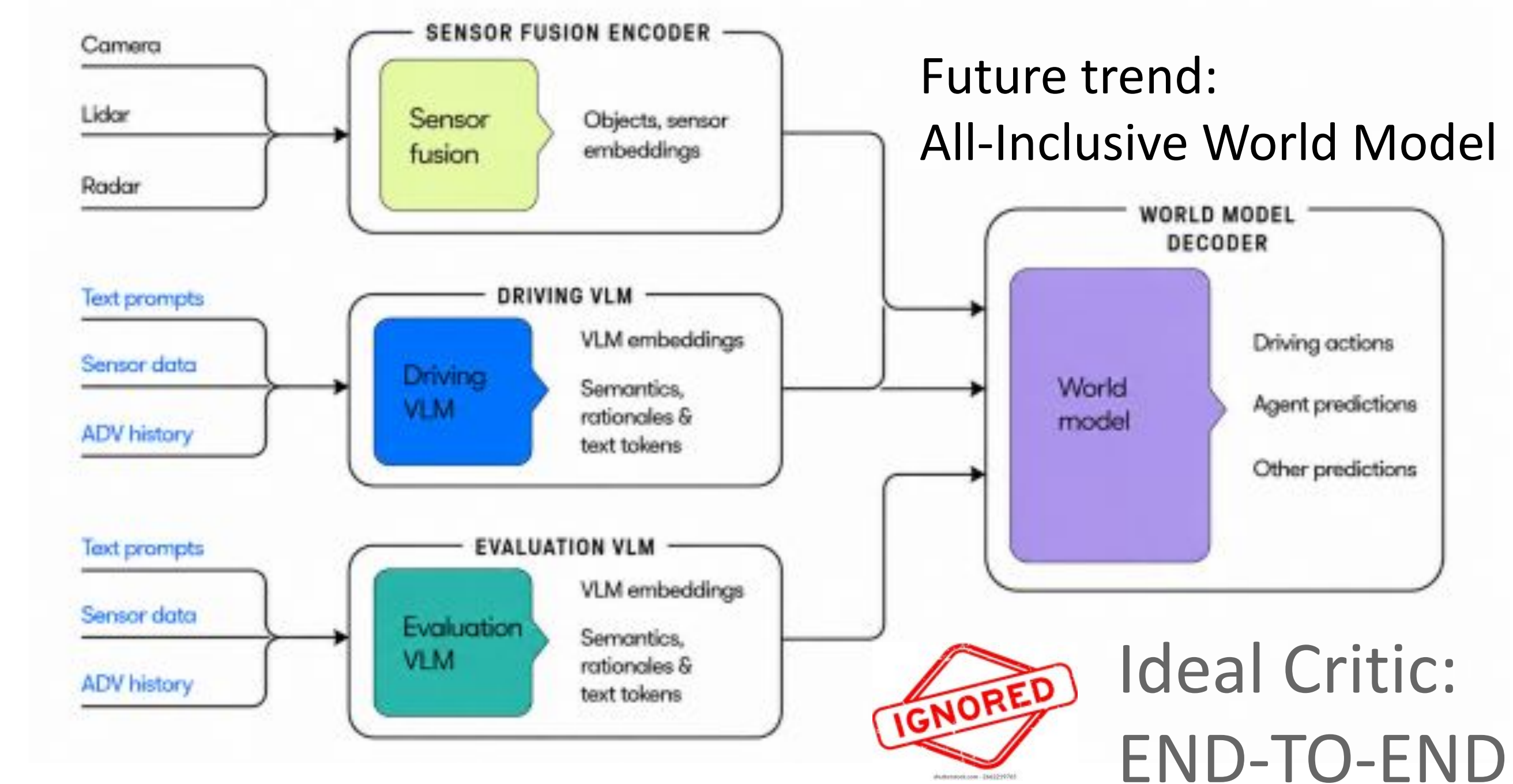
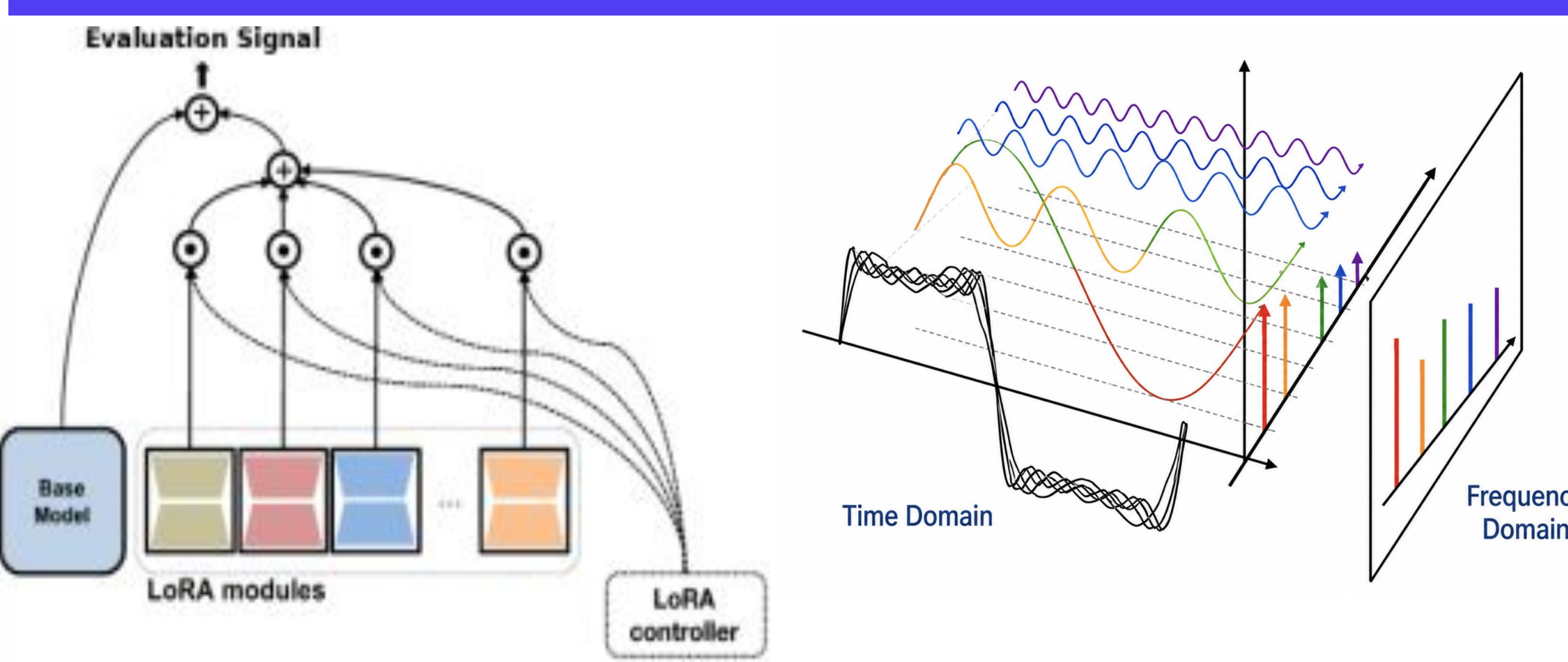


Introduction



Preliminary



MotionScorer: Motional Next Generation End-to-End Semantic Scorer

richard.chu@Motional.com

Final Deliverable: a Context-Aware Continual Evaluation Score, with Explainable Semantic Understandings

Core Research Problem: How to guide the **Score Token** (last) express the semantic understanding of Context

Contribution:

1. First to produce an **End-to-End** self-driving scoring evaluator that emits the evaluation score itself
2. introduce a new perspective of trajectory scoring: a problem of guiding and filtering its latent attention
3. a **lightweight post-training design** for trajectory evaluation.

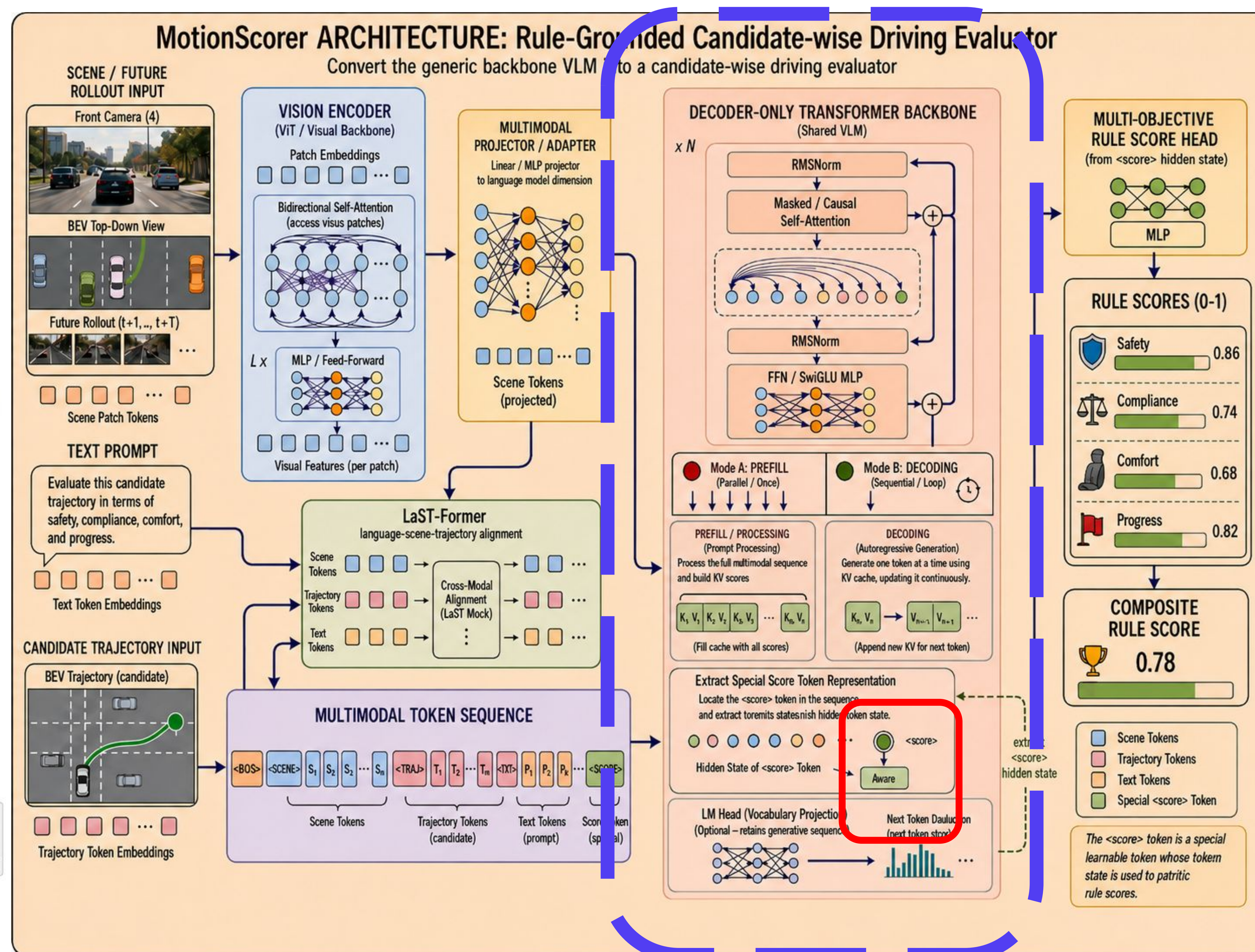
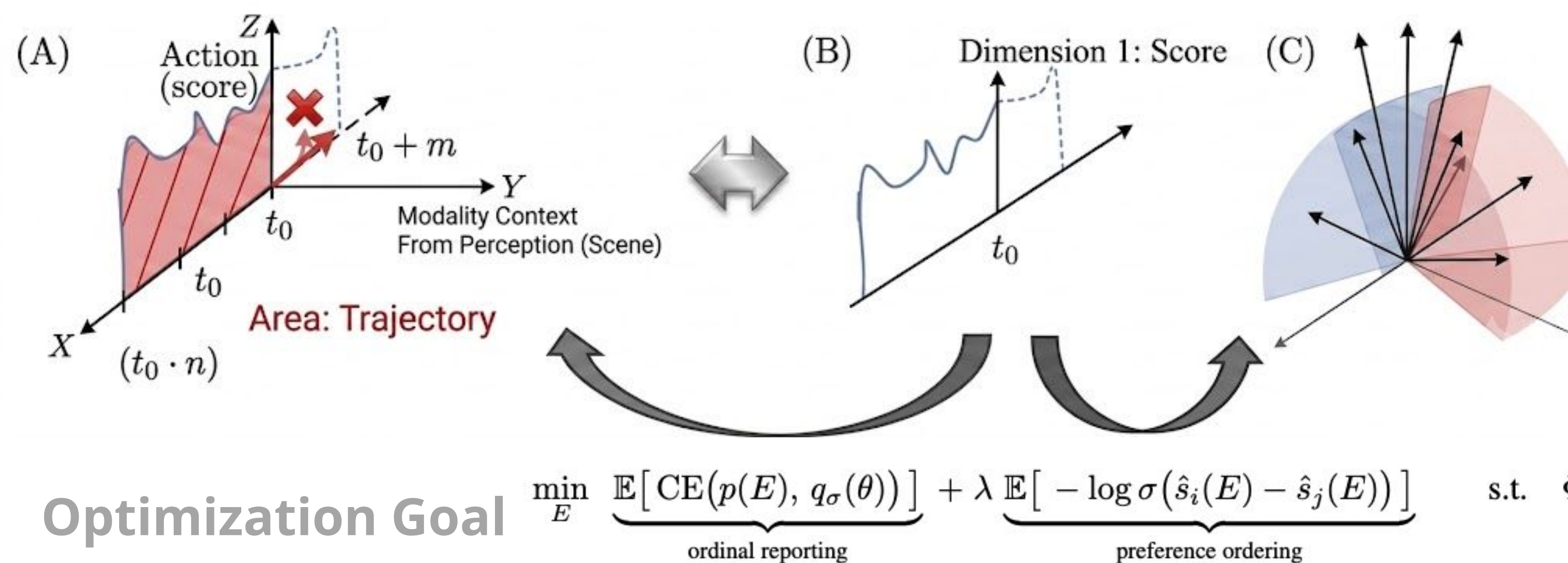


Figure 1. Main Workflow and Core Optimization Goal

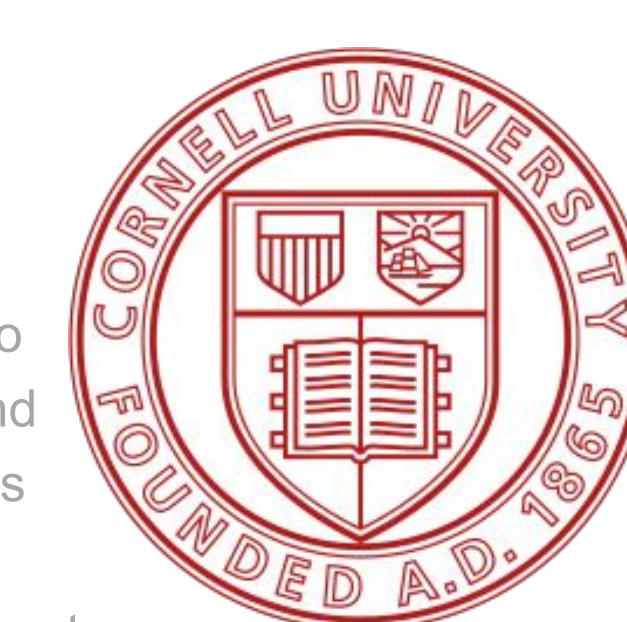
$$u^T h_q = \sum_{i=1}^n \alpha_i (\theta_i + \varepsilon_i), \quad \theta_i = \mathbb{E}[u^T v_i | \text{quality}], \quad \text{Cov}(\varepsilon) = \Sigma,$$

Optimization Problem



Acknowledgement

Worked within Motional's System Readiness & Performance team to develop data-driven evaluation and analytics pipelines for autonomous driving systems. Contributed to large-scale performance assessment.



Results

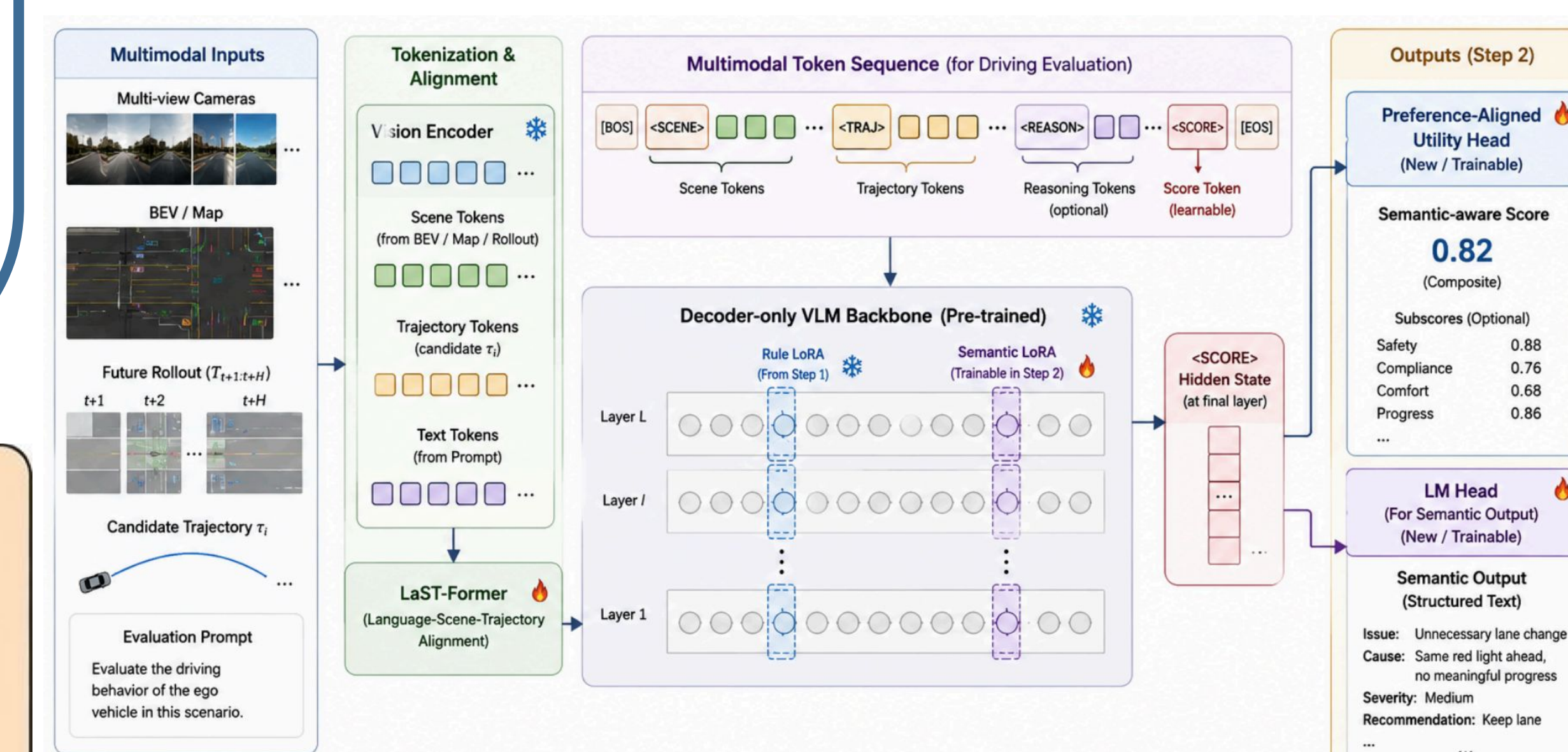


Figure 2. Multi-LoRA Details & Results

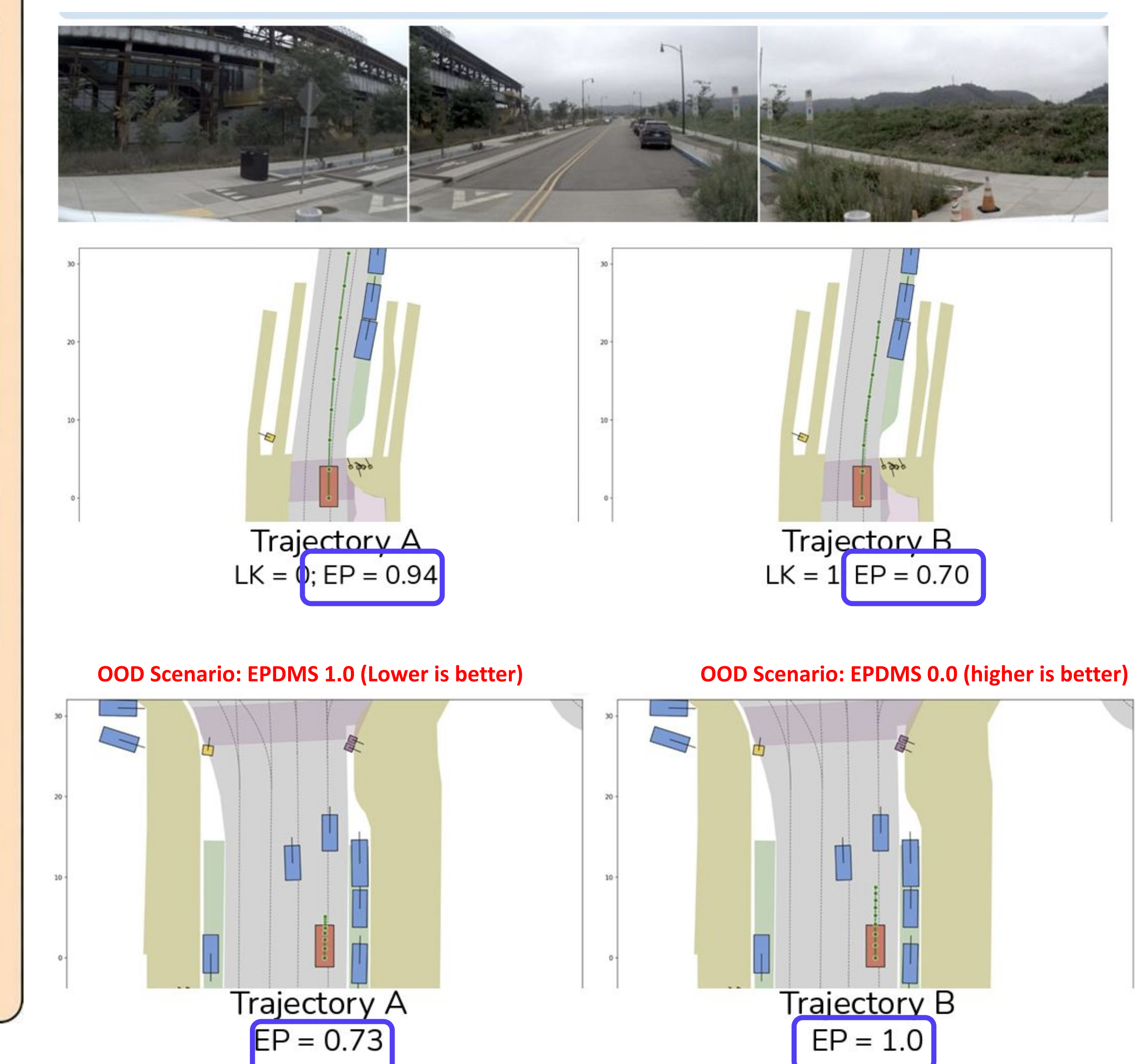


Figure 3. Highlighted Results

Method	No weight update	Absolute scalar score	Evidence-grounded rationale	Range not rule-capped
Rule-based metrics				
EPDMS / NAVSIM	✓	✓	✗	✗
SM-PDMS / StyleDrive	✓	✓	✗	△
Learned scorers / reward models				
RWM	✗	✓	✗	✗
WoTE	✗	✓	✗	△
GTRS	✗	✓	✗	△
Vision-language evaluators				
DriveJudge	✗	✓	△	✗
DriveCritic	✗	✓	✓	✓
DriveReward	✗	✓	✓	✓
VLoE	✗	✓	△	△
MotionScoring-VLM (ours)	✓	✓	✓	✓

Table 1. Effectiveness Summary

References

Lewis, Patrick, et al. "Retrieval-augmented generation for knowledge-intensive nlp tasks." Advances in Neural Information Processing Systems 33 (2020): 9459-9474.